

Iatrogenic adrenal atrophy following oral corticotherapy is not reliably identified ultrasonographically in cats

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Abstract

Background

In dogs, corticosteroid administration is known to decrease adrenal gland height when measured ultrasonographically. However, comparable information is lacking in cats.

Objectives

i) Validate that the adrenal height of our control population without corticosteroid administration was similar to previous data, ii) determine effects of dose and duration of oral corticosteroid therapy on adrenal height, and iii) determine an adrenal size threshold to differentiate cats receiving corticosteroids or not.

Animals and procedures

Adult cats ($N = 308$) that received abdominal ultrasonographic examination(s) were retrospectively recruited and allocated into 2 groups: those with and without oral corticosteroid use. Cats receiving corticosteroids were subdivided into 6 subgroups by dose (supraphysiologic, anti-inflammatory, or immunosuppressive) and duration of therapy (≤ 1 mo or > 1 mo).

Results

Adrenal height in cats without corticosteroid therapy was comparable to previous studies. An anti-inflammatory corticosteroid dose for > 1 mo caused a 21.4% decrease in adrenal height (mean difference of 0.8 mm; $P = 0.009$). However, no difference in adrenal height was noted in the other subgroups ($P > 0.21$), and no useful adrenal height threshold was established.

Conclusion and clinical relevance

Feline iatrogenic adrenal atrophy may be difficult to establish with ultrasonography, as only cats receiving anti-inflammatory corticosteroid doses for > 1 mo had a modest (< 1 mm) decrease in adrenal height.

Résumé

L'atrophie iatrogène des surrénales consécutive à une corticothérapie orale n'est pas identifiée de manière fiable par échographie chez le chat

Contexte

Chez le chien, l'administration de corticostéroïdes est connue pour diminuer la taille des glandes surrénales lorsqu'elle est mesurée par échographie. Cependant, des informations comparables manquent chez les chats.

Objectifs

i) Valider que la taille des surrénales de notre population témoin sans administration de corticostéroïdes était similaire aux données précédentes, ii) déterminer les effets de la dose et de la durée de la corticothérapie orale sur la taille des surrénales, et iii) déterminer un seuil de taille des surrénales pour différencier les chats recevant des corticostéroïdes ou pas.

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Animaux et procédures

Des chats adultes ($N = 308$) qui ont subi un ou plusieurs examens échographiques abdominaux ont été recrutés rétrospectivement et répartis en 2 groupes : ceux avec et sans corticothérapie orale. Les chats recevant des corticostéroïdes ont été subdivisés en 6 sous-groupes selon la dose (supraphysiologique, anti-inflammatoire ou immunosuppresseur) et la durée du traitement (≤ 1 mois ou > 1 mois).

Résultats

La taille des surrénales chez les chats sans corticothérapie était comparable à celle des études précédentes. Une dose de corticoïdes anti-inflammatoires pendant > 1 mois a entraîné une diminution de 21,4 % de la taille des surrénales (différence moyenne de 0,8 mm; $P = 0,009$). Cependant, aucune différence de taille surrénalienne n'a été notée dans les autres sous-groupes ($P > 0,21$) et aucun seuil de taille surrénalienne utile n'a été établi.

Conclusion et pertinence clinique

L'atrophie surrénalienne iatrogène féline peut être difficile à établir par échographie, car seuls les chats recevant des doses de corticostéroïdes anti-inflammatoires pendant > 1 mois ont présenté une diminution modeste (< 1 mm) de la taille des surrénales.

(Traduit par D^r Serge Messier)

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Introduction

Corticoid synthesis is controlled by the hypothalamic-pituitary-adrenal axis, also known as the corticotrophic axis, and it is regulated by negative feedback (1). Exogenous corticosteroids affect corticoid synthesis by the adrenal glands in dogs, humans, and cats (1,2). In dogs receiving corticosteroid therapy, there are time-related reductions in adrenal gland height secondary to corticotrophic axis suppression (1,3). For example, in healthy beagle dogs, there was a 27% decrease in ultrasonographically measured adrenal gland height after 1 mo of oral immunosuppressive glucocorticoid treatment, and a 44% decrease after 4 mo (3). Furthermore, within 1 mo after cessation of corticosteroid administration, the adrenals regained their pre-treatment size (3). The effect of glucocorticoid therapy on adrenal glands is important, as it can be a useful tool to assist the clinician in diagnosing and monitoring both iatrogenic hyper- and hypoadrenocorticism.

Despite available medical literature for the dog, little is known about effects of corticosteroid administration on ultrasonographic assessment of feline adrenal gland size. However, ultrasonographic measurements of adrenal gland size were reported in healthy cats (2,4–7), including for various weight categories (8), as well as in sick cats (5,9). Although adrenal glands of sick cats without endocrinopathies were reported to be of similar size compared to healthy cats (4,5), some feline endocrinopathies, such as hyperthyroidism and acromegaly, are associated with mild adrenomegaly (10,11).

The aims of this retrospective study were to i) validate that the ultrasonographic adrenal height of our control population of cats without corticosteroid administration was similar to previous data from the literature of cats with chronic nonadrenal diseases; ii) compare adrenal height in cats receiving oral corticosteroid therapy to those not receiving such therapy, and determine effects of both dose and duration of treatment; and iii) determine whether there is an ultrasonographic adrenal height threshold differentiating cats receiving corticosteroids from those not receiving them. Our hypotheses were that i) adrenal height of cats not receiving corticosteroids in our study population is similar to that in data from the literature; ii) adrenal height

in cats receiving corticosteroids is smaller than in cats without corticosteroids, and that this is more significant with increased doses and duration of treatment; and iii) a threshold discriminates cats without corticosteroids from cats receiving them.

Materials and methods

In this retrospective, descriptive study, all cases were recruited from 2006 to 2019 at the Centre Hospitalier Universitaire Vétérinaire of the Université de Montréal (Quebec). Inclusion criteria for the study were adult-sized cats (> 10 mo of age) that received at least 1 abdominal ultrasonographic examination. Exclusion criteria were cats with an endocrinopathy, as this can affect adrenal gland size (10–12), those with an adrenal nodule or mass, those receiving topical corticosteroids alone, those with an incomplete medical record, or those without adrenal ultrasonographic images. Case inclusion was decided by consensus among a veterinary diagnostic imaging intern (CG), an American College of Veterinary Radiology (ACVR) diplomate (CF), and an American and European College of Veterinary Internal Medicine (ACVIM — ECVIM) diplomate (BC). Signalment; adrenal size (maximal height and length of left and right adrenal glands) on sagittal plane on ultrasound images; corticotherapy type, dose and duration; and definitive or presumptive final diagnoses were retrospectively collected by a veterinary diagnostic imaging intern (CG). A diagnosis was considered presumptive when the affection positively responded to the medical treatment implemented. It was considered definitive whenever a specific diagnosis was confirmed by paraclinical tests (*e.g.*, cytology, histopathology, culture, or echocardiography). Images were acquired with 1 of 2 ultrasound machines: from 2006 to 2013, a Philips ATL HDI 5000 (Advanced Technology Laboratories, Bothell, Washington, USA); and from 2013 to 2019, a Toshiba Aplio 400 (Canon Medical System, Markham, Ontario); and were captured with either a linear transducer (5 to 12 and 13 to 18 MHz, respectively) or a convex transducer (5 to 8 and 8 to 11 MHz, respectively). Ultrasonographic examinations were completed by ACVR diplomates or ACVR residents supervised by ACVR diplomates. Patients were imaged in dorsal recumbency, abdominal hair was

Table 1. Demographics of cats receiving exogenous corticosteroids or not (CS and WCS, respectively).

	Cats receiving corticosteroids (CS)	Cats without corticosteroids (WCS)
Age, y (mean $\pm$ SD)	9.35 $\pm$ 3.53	7.96 $\pm$ 4.42
Weight, kg (mean $\pm$ SD)	4.46 $\pm$ 0.39	4.46 $\pm$ 1.45
Number of females		
Spayed	30	105
Intact	1	4
Number of males		
Neutered	34	130
Intact	1	4
Breeds	50 domestic shorthair 4 Himalayan 3 Bengal 2 Persian 1 each of Abyssinian, Oriental shorthair, sphynx, Highland fold, British shorthair, pixie-bob, Siamese	180 domestic shorthair 15 Siamese 12 Himalayan 5 Cornish rex, 5 Abyssinian 4 Persian, 4 Tonkinese 2 British shorthair, 2 Highland lynx, 2 Bengal, 2 Maine coon 1 each of Burmese, ragdoll, bleu Russe, munchkin, Selkirk rex, American shorthair, Havana brown, Devon rex, Savannah

clipped, and acoustic coupling gel was applied for the ultrasonographic examination. Depth, focus, and gain were individually adjusted according to radiologist preferences.

Recruited cats were divided into 2 groups: cats without corticosteroids (WCS) and cats with oral corticosteroid administration (CS). Oral corticosteroids included prednisolone and dexamethasone. Cats from group CS were subdivided by dose [supraphysiologic (P), anti-inflammatory (AI), and immunosuppressive (IS)] and duration of treatment (≤ 1 mo or > 1 mo). An oral supraphysiologic dose was defined as ≥ 0.2 mg/kg per day and < 1 mg/kg per day for prednisolone, and as ≥ 0.02 mg/kg per day and < 0.1 mg/kg per day for dexamethasone (1). An oral anti-inflammatory dose was defined as ≥ 1 mg/kg per day and ≤ 2 mg/kg per day for prednisolone, and as ≥ 0.1 mg/kg per day and ≤ 0.2 mg/kg per day for dexamethasone (1). Finally, an oral immunosuppressive dose was defined as > 2 mg/kg per day for prednisolone, and as > 0.2 mg/kg per day for dexamethasone (1,2). Corticosteroid doses in the subgroups were based on published recommendations (1,8,13). Cats from group CS that had an ultrasonographic examination before corticosteroid treatment implementation (Time 0) were subclassified as CS at Time 0. For cats not receiving corticosteroids (group WCS and group CS at Time 0) that had > 1 ultrasound examination, adrenal dimensions were collected from the initial examination. For cats receiving corticosteroids (group CS after Time 0 or without Time 0), adrenal dimensions were collected from the ultrasound examination done when the cat received the highest corticosteroid dose.

Statistical analyses were completed by a statistician (GB) using commercial graphing and statistical analysis software

Table 2. Definitive and presumptive final diagnoses of cats given exogenous corticosteroids or not (CS and WCS, respectively).

Diseases	Number of cats	
	Cats receiving corticosteroids (CS)	Cats without corticosteroids (WCS)
Hepatobiliary and digestive	47	155
Neoplasia	13	81
Myelopathies	2	9
Cardiorespiratory	9	34
Infectious	10	13
Urinary tract	15	52
Hematopoietic and thrombosis	10	27
Dermatology	—	2
Fever of unknown origin	—	3
Miscellaneous	—	8

The Miscellaneous category included 1 case of trauma, 1 case of autoimmune disease (systemic lupus erythematosus), 4 cases of idiopathic hypercalcemia, and 2 cases of abdominal hernia. Note that this table lists all final diagnoses in the study population, and that a given cat can have ≥ 1 disease presented here and thus be counted in the > 1 category (see Table 3).

Table 3. Number of concurrent diseases per cat in cats given exogenous corticosteroids or not (CS and WCS, respectively).

Number of diseases per cat	Cats receiving corticosteroids (CS)	Cats without corticosteroids (WCS)
1	33	126
2	20	78
3	10	27
4	3	8
5	—	3
6	—	1

Table 4. Number of data recorded (adrenal height and length), bilaterally or unilaterally, in cats given exogenous corticosteroids or not (CS and WCS, respectively).

Data recorded	Cats receiving corticosteroids (CS) (n = 66)		Cats without corticosteroids (WCS) (n = 242)	
	Adrenal height	Adrenal length	Adrenal height	Adrenal length
Bilaterally	52	52	195	194
Right only	5	5	18	19
Left only	9	9	29	29
Total	66	65	242	242

Note the "n" indicates the total number of cats in each group.

(SAS v.9.4; SAS Institute, Cary, North Carolina, USA). When needed, average adrenal height was calculated for each individual using data from both sides. A Student's *t*-test for unequal variances was used to compare adrenal size in groups WCS and CS Time 0. A 95% confidence interval about the mean was used to compare groups WCS and CS Time 0 with the mean reported in the literature (5). A linear model was used to compare adrenal size in various treatments groups, followed by Dunnett's *post hoc* tests to compare the mean in group WCS and the mean in other groups. A linear mixed model was used to assess influences of treatment duration and dose on adrenal measurements in groups with data from the same individuals at various intervals. Finally, an ROC curve was established to determine an adrenal height threshold that would combine maximal sensitivity and specificity to distinguish cats from groups WCS and CS. Results are

Table 5. Feline adrenal size in mm (mean $\pm$ SD) in our study^a and in previous literature^b (5).

	Left adrenal gland		Right adrenal gland		Left and right adrenal height	Left and right adrenal length
	Height	Length	Height	Length		
Cats without endocrinopathies (WCS) ^a	3.8 $\pm$ 0.8	9.3 $\pm$ 2.1	3.9 $\pm$ 0.9	9.3 $\pm$ 2.4	3.9 $\pm$ 0.7	9.3 $\pm$ 1.8
Cats before corticosteroid treatment (CS at Time 0) ^a	4.0 $\pm$ 0.8	9.3 $\pm$ 3.1	4.1 $\pm$ 0.9	9.5 $\pm$ 2.7	4.0 $\pm$ 0.7	9.3 $\pm$ 2.3
Healthy cats ^b	3.6 $\pm$ 0.7	10.4 $\pm$ 1.8	3.6 $\pm$ 0.7	10.8 $\pm$ 1.9	—	—
Chronically sick cats (without endocrinopathies) ^b	3.9 $\pm$ 0.8	10.8 $\pm$ 2.0	3.9 $\pm$ 1.0	10.9 $\pm$ 2.0	—	—

In the previous literature, chronically sick cats had nonadrenal-related diseases and did not have any endocrinopathy.

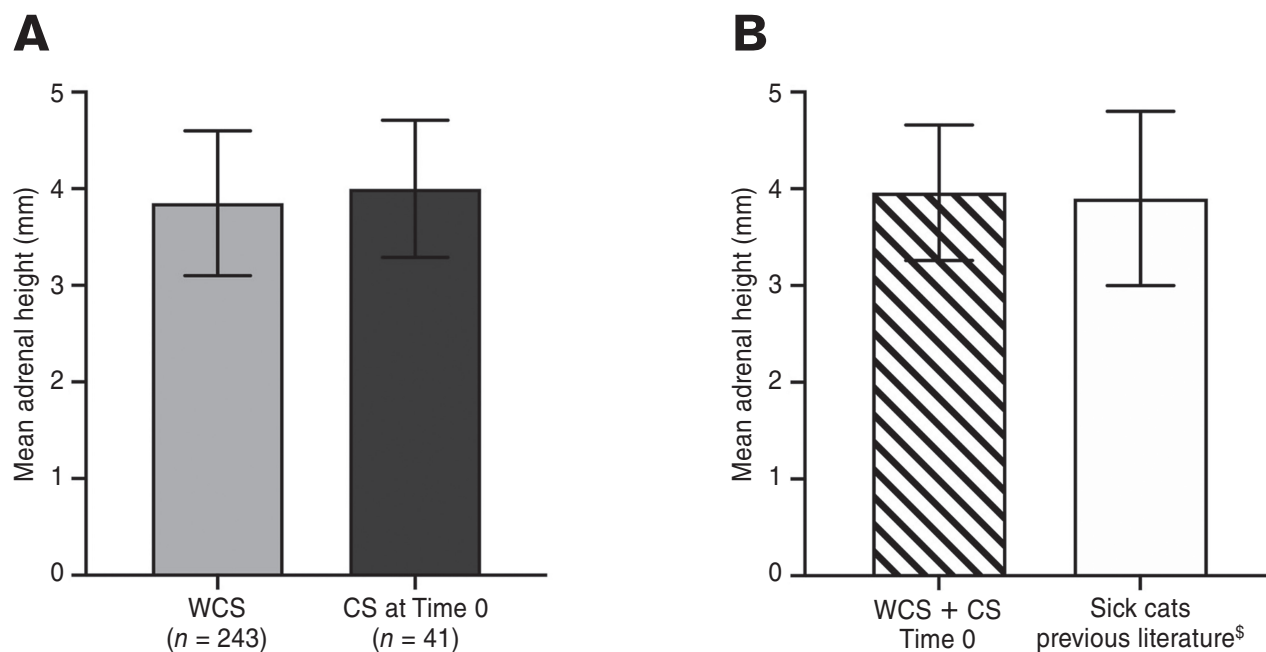


Figure 1. A – Comparison of mean adrenal height in our study cats without corticosteroid treatment (WCS) and before starting corticosteroid treatment (CS at Time 0). There was no significant difference in adrenal height between the 2 groups. B – Comparison of mean adrenal height in cats without corticosteroid in our study with published data[§] (5). There was no significant difference between the 2 populations. Bars indicate standard deviation (SD).

expressed as mean $\pm$ standard deviation unless stated otherwise. Statistical significance was set at $P < 0.05$.

Results

There were 373 cats initially recruited, but 65 were excluded. Therefore, a total of 308 cats with 512 ultrasound examinations were included in this retrospective study. Of these, 242 cats did not receive corticosteroids (group WCS) and 66 cats received oral corticosteroids (group CS). Of the 66 cats in group CS, 41 had an ultrasonographic examination before corticosteroid treatment (Time 0). The remainder of the demographic data for all patients is detailed in Table 1.

Definitive or presumptive final diagnoses of the 308 included cats are listed in Table 2, whereas concurrent pathologies are listed in Table 3. In group CS, 50% of the cats were diagnosed with only 1 disease and 50% were diagnosed with > 1 disease. In group WCS, 51.8% of the cats were diagnosed with only

1 disease and 48.1% had > 1 disease, with up to a maximum of 6 comorbidities in 1 patient.

The numbers of adrenal gland measurements recorded (length and height, bilaterally or only on the right or on the left) in the 2 groups of cats are shown in Table 4.

There were no significant differences between cats in groups WCS and CS for sex, age, or weight.

There were no differences between groups WCS and CS at Time 0 for mean adrenal height and length on the right and on the left sides ($P = 0.22$ and $P = 0.96$, respectively) (Table 5 and Figure 1 A). The 95% confidence interval about the mean adrenal height of all cats without corticosteroid therapy in our study (3.87 to 4.05 mm) included the previously reported mean adrenal height in cats with chronic nonadrenal diseases (3.9 $\pm$ 0.9 mm) (Figure 1 B) (5). The 95% confidence interval for mean adrenal length of all cats without corticosteroids in our study was 9.06 to 9.54 mm (Table 5) (5).

Table 6. Distribution of cats according to oral corticosteroid treatment (prednisolone or dexamethasone) and subgroups in our study.

	Group P		Group AI		Group IS		Total
	< 1 mo	> 1 mo	< 1 mo	> 1 mo	< 1 mo	> 1 mo	
Prednisolone	12	24	8	9	4	4	61
Dexamethasone	—	4	—	—	1	—	5
Total	40		17		9		66

AI — Anti-inflammatory dose (prednisolone ≥ 1 mg/kg per day and ≤ 2 mg/kg per day, or dexamethasone ≥ 0.1 mg/kg per day and ≤ 0.2 mg/kg per day); IS — Immunosuppressive dose (prednisolone > 2 mg/kg per day, or dexamethasone > 0.2 mg/kg per day); P — Supraphysiologic dose (prednisolone ≥ 0.2 mg/kg per day and < 1 mg/kg per day, or dexamethasone ≥ 0.02 mg/kg per day and < 0.1 mg/kg per day).

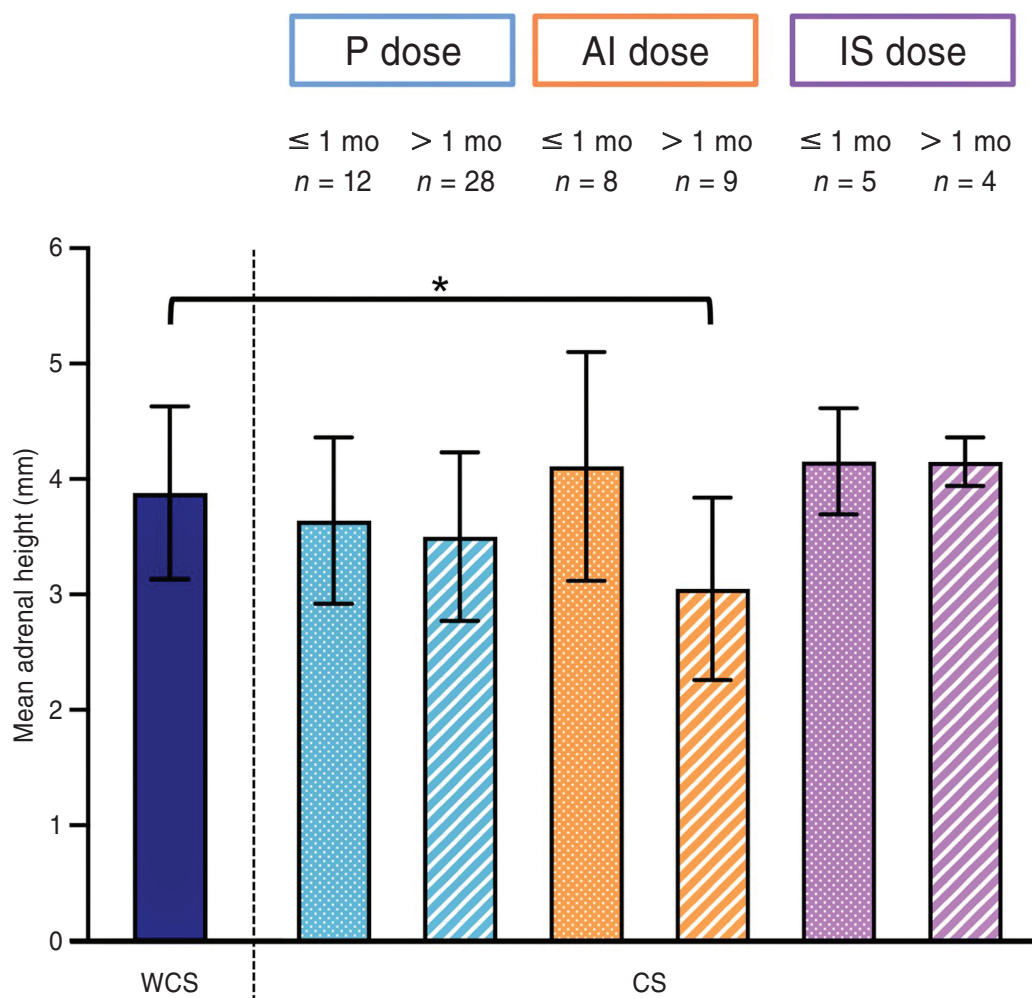


Figure 2. Comparison of mean adrenal height in cats in the 6 subgroups of the corticosteroid group (CS) and in the group that did not receive corticosteroids (WCS). Only the subgroup that received an anti-inflammatory dose of corticosteroids (AI) for > 1 mo had a significantly smaller height compared to the WCS group. Bars indicate standard deviation (SD). * $P < 0.05$.

AI – Anti-inflammatory dose; IS – Immunosuppressive dose; P – Supraphysiologic dose.

Of the 66 cats in group CS, 12 cats were assigned to subgroup P ≤ 1 mo, 28 cats to subgroup P > 1 mo, 8 cats to subgroup AI ≤ 1 mo, 9 cats to subgroup AI > 1 mo, 5 cats to subgroup IS ≤ 1 mo, and 4 cats to subgroup IS > 1 mo (Table 6). Of these cats, 61 cats received oral prednisolone and 5 received oral dexamethasone (Table 6). Adrenal dimensions in these 6 subgroups were compared to those in group WCS. Subgroup AI > 1 mo had a decrease of 21.4% in mean adre-

nal height (mean difference of 0.8 mm, $P = 0.009$; Figure 2). However, there were no differences for mean adrenal height of other subgroups ($P > 0.21$), nor for mean adrenal length in any subgroup ($P = 0.73$).

In group CS, 41 cats had an ultrasonographic examination at Time 0, with their follow-up examinations compared to this baseline. Subgroups AI and IS were grouped together (AI+IS) for statistical analyses due to low subject numbers. There was

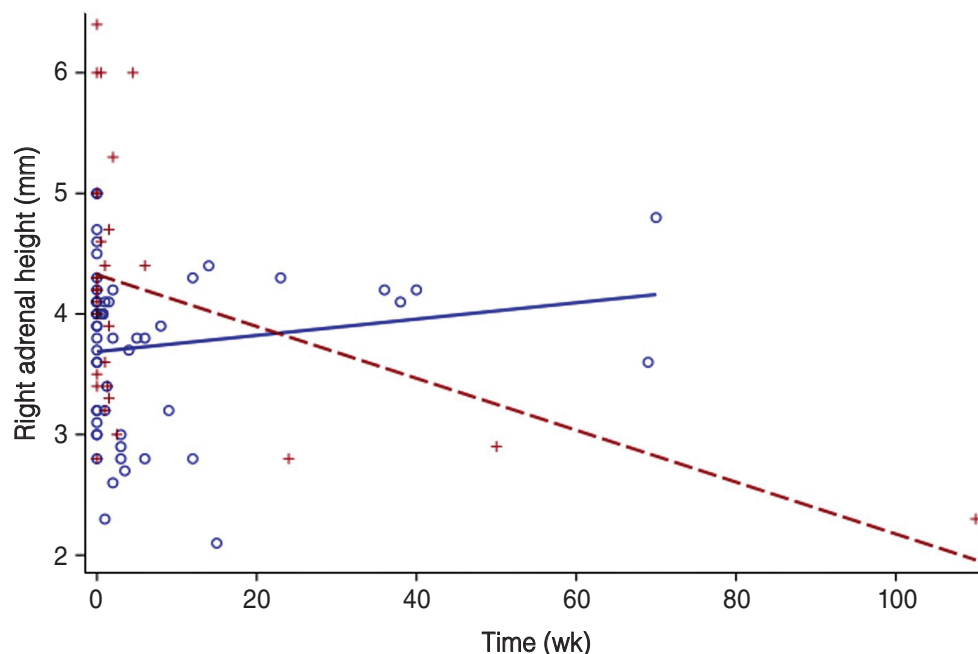


Figure 3. Mixed linear model of the evolution over time of the right adrenal gland height of cats receiving corticosteroids at supraphysiologic doses (circles and solid line) or combined anti-inflammatory and immunosuppressive doses (crosses and dashed line) when compared to their own Time 0. Circles and crosses represent individuals, whereas solid and dashed lines represent linear regression lines. A significant decrease over time for the right adrenal height is present at combined anti-inflammatory and immunosuppressive dose (AI+IS).

a decrease in the right adrenal height of subgroup AI+IS as a function of time ($P = 0.033$) (Figure 3). However, there was no difference for right adrenal gland height at dose P ($P = 0.34$). Similarly, there were no differences for the left adrenal height or mean adrenal height (left and right) over time at dose P, or in subgroup AI+IS when compared to Time 0 ($P = 0.16$ and $P = 0.18$, respectively). Finally, no differences were present for the right, left, or mean adrenal lengths at any dose ($P = 0.70$, $P = 0.26$, and $P = 0.60$, respectively).

An ROC curve analysis established an optimal adrenal height threshold of 3.85 mm to distinguish cats from groups CS and WCS, with a sensitivity and specificity of 52% (Figure 4). For a sensitivity of 100%, the adrenal height threshold was 1.8 mm (*i.e.*, all cats with an adrenal height < 1.8 mm received corticosteroid therapy).

Discussion

In this feline population, a decrease in mean adrenal height was only noted in the subgroup of cats receiving corticosteroids at an anti-inflammatory dose for > 1 mo; however, the decrease was marginal (< 1 mm). Such a small ultrasonographic measure is prone to operator and computed-measurement error (14). Thus, in the authors' opinion, its clinical usefulness remains uncertain. Moreover, adrenal size in that subgroup substantially overlapped with measurements in the group of cats that did not receive corticosteroids (group WCS), and with previously reported measurements in cats with chronic nonadrenal diseases that did not receive corticosteroids (5). These results refuted our second hypothesis and contrast with what is reported for dogs receiving hydrocortisone, in which glucocorticosteroids

induced ultrasonographically measured adrenal atrophy (1,3). In this study, the majority of cats in group CS received prednisolone (61 cats), and a minority received dexamethasone (5 cats) (Table 6). However, both of these molecules are known to suppress the hypothalamic-pituitary-adrenal axis in cats and dogs, as does hydrocortisone (1,15). Thus, the ultrasonographic differences between the cat and the dog were most likely not attributable to the class of glucocorticoid. A plausible explanation for these differences could be that cats are less sensitive to corticosteroids compared to dogs (1). There are fewer corticosteroid receptors in cats than in dogs, and they have lower affinity to corticoids (1,15). This could explain why measurable iatrogenic adrenal atrophy in the cat is not easily identified ultrasonographically.

Adrenal height in cats without corticosteroids (groups WCS and CS at Time 0) from our population was similar to that in previous published studies, and smaller than in dogs (5,6,16). It is interesting to note that the 2 feline populations compared are from North America and Europe, respectively, with similarities between 2 populations of cats from different continents. Smaller adrenal glands in cats compared to dogs may render size variation more difficult to detect ultrasonographically. However, it remains possible that a more pronounced feline adrenal atrophy would be noted ultrasonographically with a longer duration of therapy or with higher corticosteroid doses than were studied here.

Only a significant decrease in the right adrenal gland height was present over time at combined AI and IS doses (AI+IS) when compared to baseline at Time 0, but there was no significant difference over time for the left adrenal gland. This was

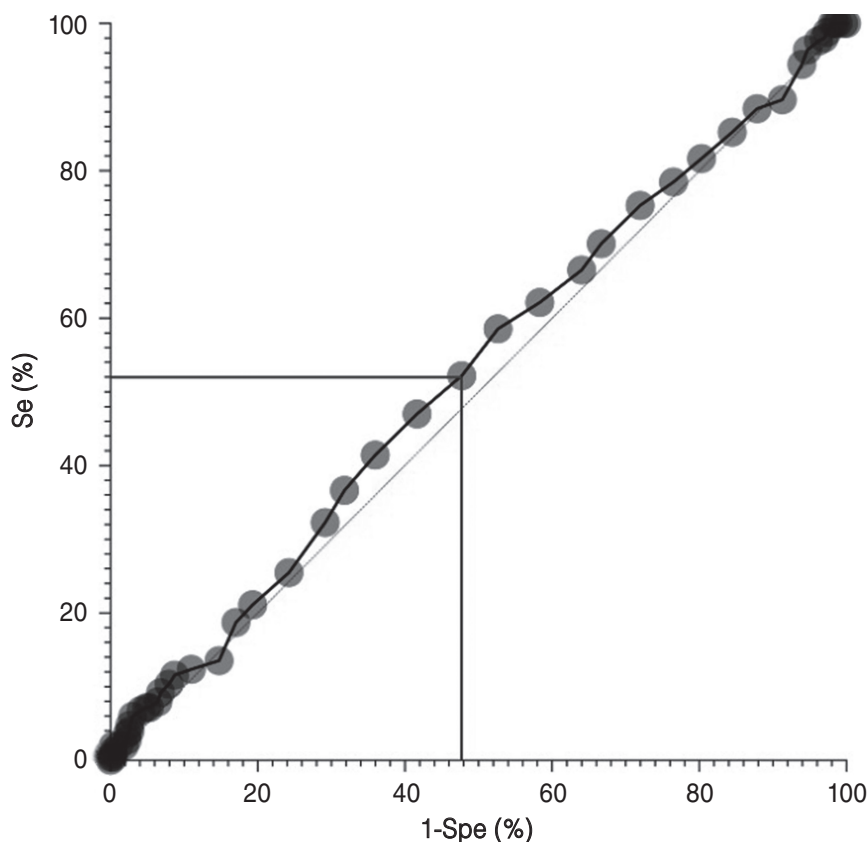


Figure 4. ROC curve analysis. At a combined maximal sensitivity (Se) and specificity (Spe) of 52%, adrenal height threshold between cats receiving corticosteroid treatment and cats without corticosteroid treatment was 3.85 mm.

surprising since, as previous studies have not reported any difference in adrenal size per side in healthy cats and cats with chronic diseases (5,9,17), a decrease in adrenal height would logically be expected to be bilateral and not unilateral. In addition, there were no significant differences between the mean heights of the right and the left adrenals at doses AI and IS, or at any therapy duration time point. Also, there was no significant difference for the mean height of right *versus* left adrenal over time. More left adrenal glands were retrospectively available for review than right adrenal glands, which could have induced a bias. Perhaps there was a degree of obliquity in some of the images, resulting in some of the recorded images not representing the actual maximal dimension of the organ; however, this should not have influenced measurements over time. Therefore, the explanation for this unilateral decrease in adrenal height over time at anti-inflammatory and immunosuppressive corticosteroids doses remains unknown. Additional prospective studies evaluating the 3D volumetric size of the feline adrenal glands would be helpful to further assess this finding.

The adrenal height threshold of 3.85 mm with an optimal sensitivity and specificity differentiating cats receiving corticosteroids from cats without corticosteroids established by the ROC curve was within the range of normal feline adrenal gland size (3.9 ± 0.7 mm in group WCS). Therefore, this threshold cannot be clinically used to distinguish cats receiving corticosteroids from those not receiving them, and our third hypothesis

was refuted. Also, the threshold of 1.8 mm at a sensitivity of 100% did not seem clinically relevant, as only 2 individuals in group CS had an adrenal height < 1.8 mm. This threshold was too low to encompass the majority of cats receiving corticosteroids. Therefore, no useful clinical ultrasonographic threshold could be established.

Adrenal length did not contribute any additional, pertinent information. There was no significant difference between adrenal lengths of groups WCS and CS. For adrenal length, there was no effect of dose or duration of corticosteroid treatment in all CS subgroups, and no statistical change over time when compared to Time 0. In previous studies, adrenal length was not correlated with body weight, body surface area, or body condition score (9); and varied slightly, but with no clinical relevance, between healthy cats and cats with nonadrenal diseases (5,9). Also, this ultrasonographic measurement had higher variability within and between observers compared to adrenal height in sagittal and transverse planes (4,17). Thus, it is usually not the preferred measurement to quantify adrenal size in practice. Furthermore, the 95% confidence interval for adrenal length of all cats without corticosteroids in our study was smaller than in previous data from the literature (Table 5) (5). This can be explained by the higher variability of this measurement. Also, maximal length is not routinely measured at our hospital, so images retrospectively available were not optimized for adrenal length (5).

This study has several limitations, mainly associated with its retrospective nature. Ultrasonographic examinations were done at various time points after corticosteroid treatment implementation. Although there were 2 subgroups of treatment duration ($\leq$ and > 1 mo), there were large differences in duration in each group, and this temporal heterogeneity may have added a bias. Corticosteroid treatment protocols were not standardized because they were adjusted based on clinician recommendation at the time of the diagnosis. Included cats received prednisolone or dexamethasone but, as mentioned above, both of these molecules suppress the hypothalamic-pituitary axis; a decision was thus made to include all cats. There was no attempt to compare the influence of prednisolone *versus* dexamethasone, as this was not the focus of our study and it would have been difficult given the small number of individuals receiving dexamethasone. Another limitation is that limited numbers of individuals received high corticosteroid doses or long durations of treatment, reducing statistical power in these subgroups. In addition, various operators completed the ultrasound examinations without any standardized protocol, and no videos of the adrenal glands were available for review (only static images). Thus, some measurements may have been underestimated due to obliquity. This issue likely predominantly affected adrenal length measurements but had minimal effect on adrenal thickness, as this is routinely assessed ultrasonographically and images should be optimized to capture it. Other ultrasonographic adrenal characteristics (form, echogenicity, mineralization) were not evaluated because they were outside the scope of this study. Finally, no ACTH stimulation test was done during the follow-up ultrasound for cats in group CS; therefore, the degree of corticotrophic axis suppression was not confirmed. Based on previous studies, suppression was expected at the doses used here (1,15). Further prospective studies would be needed to analyze higher doses and longer durations of oral corticosteroid treatment with control of the corticotrophic axis at the time of the ultrasound examinations, and with standardized corticosteroid administration protocols.

In conclusion, although a decrease in adrenal height was noted in the subgroup of cats receiving anti-inflammatory doses of oral corticosteroids for a duration of > 1 mo, the change was marginal (< 1 mm), and overlapped with adrenal height in cats that did not receive oral corticotherapy. Therefore, iatrogenic adrenal atrophy in cats at the doses and durations of therapy studied was not reliably detected with ultrasonography.

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